

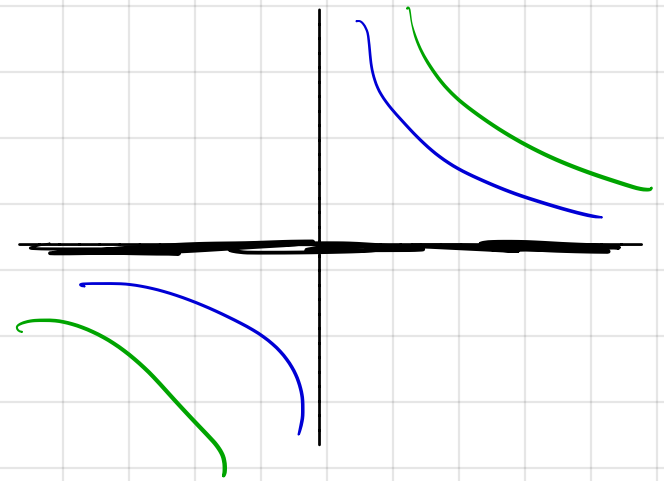
$$1) f(x, y) = \sqrt{xy}$$

Domain

$$\{(x, y) \in \mathbb{R}^2 \mid xy \geq 0\} = (-\infty, 0] \times (-\infty, 0] \cup [0, \infty) \times [0, \infty)$$

$$f(x, y) = c \quad \sqrt{xy} = c$$

$$y = \frac{c^2}{x}$$



$$- y = 0$$

$$- y = \frac{1}{x}$$

$$- y = \frac{c}{x}$$

$$2) \lim_{(x, y) \rightarrow (1, 1)} \frac{\sqrt{xy}}{x^2 + y^2} = \frac{\sqrt{1 \cdot 1}}{1^2 + 1^2} = \frac{1}{2}$$

10.2

$$16) \lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 - y^2}{x^2 + y^2} = \frac{3 \cdot 0^2 - 0^2}{0^2 + 0^2} = \frac{0}{0} \text{ indeterminate}$$

① positive  $x$ -axis:  $y=0$

$$\lim_{x \rightarrow 0^+} \frac{3x^2 - 0^2}{x^2 + 0^2} =$$

$$\lim_{x \rightarrow 0^+} 3 = 3$$

② positive  $y$ -axis:  $x=0$

$$\lim_{y \rightarrow 0^+} \frac{3 \cdot 0^2 - y^2}{0^2 + y^2} =$$

$$\lim_{y \rightarrow 0^+} -1 = -1$$

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y)$$

1) Evaluate  $f(a,b)$

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, 1^{\infty}, \infty - \infty, 0^{\infty}, \infty^0 \leftarrow$  Indeterminate

2) Check along directions, e.g.  $y = g(x), x = h(y)$

$$\lim_{x \rightarrow a} f(x, g(x)) \text{ OR } \lim_{y \rightarrow b} f(h(y), y)$$

3) Compare limits

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = \text{DNE}$$

10.2

$$a) \lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^3 + yx}$$

$$\textcircled{1} y = mx$$

$$\lim_{x \rightarrow 0} \frac{2x(mx)}{x^3 + (mx)x} =$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x^2} \cdot 2m}{\cancel{x^2} \cdot (x+m)} =$$

$$\lim_{x \rightarrow 0} \frac{2m}{x+m} = \frac{2m}{0+m} = 2$$

$$\textcircled{2} y = x^2$$

$$\lim_{x \rightarrow 0} \frac{2x \cdot x^2}{x^3 + x^2 x} =$$

$$\lim_{x \rightarrow 0} \frac{2x^3}{2x^3} = 1$$

\*  
DNE

# Partial Derivatives $f(x, y)$

Notation

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial x} = f_x = f_x(x, y)$$

$$\frac{\partial f}{\partial y} = f_y$$

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$\frac{\partial^2 f}{\partial y^2} = f_{yy}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) = f_{yx} = (f_y)_x$$

$$f_{xy} = f_{yx}$$

10,3

$$3) f(x, y) = (xy)^{3/2} - (xy)^{2/3}$$

$$\frac{\partial f}{\partial x} = \frac{3}{2}(xy)^{1/2}y - \frac{2}{3}(xy)^{-1/3}y$$

$$\frac{\partial}{\partial x}(xy) = y$$

$$\frac{\partial f}{\partial y} = \frac{3}{2}(xy)^{1/2}x - \frac{2}{3}(xy)^{-1/3}x$$

$$13) f(x, y) = \ln(2x + y)$$

$$\frac{\partial f}{\partial x} = \frac{1}{2x + y} \cdot 2$$

$$\frac{\partial f}{\partial y} = \frac{1}{2x + y}$$

$$38) f(x, y, z) = y \tan(x^2 + z)$$

$$\frac{\partial f}{\partial x} = y \sec^2(x^2 + z) \cdot 2x$$

$$\frac{\partial f}{\partial y} = \tan(x^2 + z)$$

$$\frac{\partial f}{\partial z} = y \sec^2(x^2 + z) \cdot 1$$

$$43) f(u, w) = \tan(u + w)$$

$$\frac{\partial^2 f}{\partial u^2} = \frac{\partial}{\partial u} \left( \frac{\partial f}{\partial u} \right)$$

$$\frac{\partial f}{\partial u} = \sec^2(u + w)$$

$$\frac{\partial^2 f}{\partial u^2} = 2 \sec(u + w) \cdot \sec(u + w) \tan(u + w)$$

$$\tilde{4b) f(x, y) = e^{-x^2 - y}$$

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) \right)$$

$$\frac{\partial f}{\partial x} = -2x e^{-x^2 - y}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x} &= -2x e^{-x^2 - y} \cdot (-1) = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) \\ &= 2x e^{-x^2 - y} \end{aligned}$$

$$\begin{aligned} \frac{\partial^3 f}{\partial^2 y \partial x} &= 2x e^{-x^2 - y} \cdot (-1) = \frac{\partial}{\partial y} \left( \frac{\partial^2 f}{\partial y \partial x} \right) \\ &= -2x e^{-x^2 - y} \end{aligned}$$